

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (CE/IT)

March-Apr 2018

IT 201 Database Management System

Date: 02.04.2018, Monday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q – 1 Answer the following questions. [07]

(a) A table in DBMS can have only one [01]

- (a) Secondary Key (b) Alternate Key
(c) Unique Key (d) Primary Key

(b) Define the term: Schema and Instance. [02]

(c) Draw and explain Physical, Logical and View level architecture of DBMS. [02]

(d) List down the different types of query languages. [02]

Q - 2 (a) Construct an E-R diagram for banking system. [05]

(b) Explain specialization and generalization with appropriate example. [05]

(c) What is the role of Database administrator in DBMS? Explain in detail. [04]

OR

(c) List and explain any four basic operator of relational algebra in detail. [04]

Q – 3 Answer the following questions. [Any TWO] [14]

(a) What is Normalization? Explain 1NF, 2NF, 3NF in detail with example.

(b) Explain the disadvantages of file processing system over database system in detail.

(c) Define functional dependency. Explain the following armstrong's axioms:
Reflexivity, Augmentation, Transitivity, Union and Decomposition.

SECTION – II

Q – 4 Answer the following questions. [07]

(a) The _____ set operator combines all tuples from two relations. [01]

Select B) Project C) Union D) Rename

(b) What is the role of commit and rollback command? [02]

(c) Differentiate between primary key and candidate key. [02]

(d) Explain shared-lock and exclusive-lock of concurrency control scheme. [02]

Q - 5 (a) Explain the wait-die and wound-wait scheme of dead lock in detail. **[05]**

OR

Q - 5 (a) List and explain the ACID property of Transaction. **[05]**

(b) Explain the Conflict serializability and View serializability of transaction in detail. **[09]**

Q - 6 **Answer the following questions. [Any TWO]** **[14]**

(a) Explain the two-phase commit protocol in detail.

(b) Describe log based recovery technique of recovery system of DBMS.

(c) Briefly describe the steps involved in query processing.
